

Reliable attitude control integrating reaction wheels and embedded asymmetric magnetorquers for detumbling CubeSats

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Abstract

The advent of CubeSats, compact satellites with a standardized form factor, has disrupted traditional space research paradigms, fostering innovation and collaboration across academic institutions and industries. Following deployment, CubeSats typically employ magnetorquer rods to initiate the detumbling sequence, gradually reducing the angular velocity before transferring control to reaction wheels for complete spin neutralization. However, this conventional approach entails a substantial spacecraft space requirement, necessitating an alternative and disruptive methodology. Furthermore, considering that approximately 50% of Attitude Determination and Control Subsystems (ADCS) failures are attributed to faults related to the moving parts in reaction wheels, strategies are taken into consideration to address a worst-case scenario of reaction wheels failure. This paper introduces a disruptive approach that uses diverse geometries and a non-unity track width ratio in PCB-integrated magnetorquers with reaction wheels for comprehensive control. We demonstrate the effectiveness of various coil configurations through a series of extensive simulations and establish a systematic framework to select optimal hybrid designs tailored to specific mission requirements.

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1. Introduction

The advent of small satellites can be traced back to the mid-20th century when the concept of miniaturized satellites was first explored. However, it wasn't until the 1990s that the term “CubeSat” was coined, marking a significant milestone in the development of small satellite technology. Small satellites have revolutionized space exploration and

satellite technology, embodying a paradigm shift with their innovative and transformative capabilities (Xu et al., 2016). These remarkable engineering marvels, referred to as CubeSats or nanosatellites, have irresistibly captured global attention due to their compact footprints, expedited development timelines, and cost-effective solutions showcasing their compelling appeal. Their importance lies in their ability to democratize access to space, allowing researchers,

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Nomenclature

Variable Description

p, q, r	Angular velocities in the body frame of the vehicle.
q_0, q_1, q_2, q_3	Quaternions.
ϕ, θ, ψ	Euler roll, pitch, and yaw angles
m	Mass
$\hat{n}_{Ri}$	The axis about which the reaction wheel rotates.
$\vec{r}_{Ri}$	The distance from the center of mass of the spacecraft to the center of mass of the reaction wheel in the vehicle body reference frame.
$\vec{\omega}_{B/I}$	Angular velocity vector of the satellite in the body frame.
I	Mass moment inertia matrix about the mass center in the body frame.
T_{IB}	Rotation matrix from frame I to frame B
S	Skew symmetric operator on a vector. Multiplying it with a vector results in a cross product.
I_s	Change in inertia in the body frame caused by deployment of solar panels and/or antenna.
ϵ	Denotes roll ϕ , pitch θ or yaw ψ depending on which reaction wheel is being used
L, M, N	Components of the total moment applied to CubeSat in body frame.
i_x, i_y, i_z	Applied current in each magnetorquer.

$\vec{M}_M$	Torque produced by the magnetorquers.
$\vec{H}_S$	The total angular momentum of the vehicle.
$\vec{M}_{desired}$	The desired torque to be placed on the satellite.
J	$3 \times N_{RW}$ matrix which correlates mass moment of inertia and orientation of reaction wheels.
M_x	Number of magnetorquer coils in desired topological reconfigurable arrangement.
$\vec{D}_0$	Magnetic moment of a single coil.
P_0	Single coil's power consumption.
R_0	Single coil's Resistance.
$T_{external}$	The effective temperature resulting from the external environment.
$T_{internal}$	The total thermal heat experienced by the cube-sat due to the external environment including the planar coils.
σ	Stefan-Boltzmann constant for radiative heat transfer calculations.
F	Fill factor/emissivity factor in the Stefan-Boltzmann equation, assessing surface efficiency in thermal radiation emission.
q_s	Parameter for evaluating radiative heat transfer utilizing incoming solar energy.

educational institutions, and emerging space players to actively participate in the extraterrestrial frontier (Sweeting, 2018). Small satellites have ushered in a new era by breaking free from the limitations of traditional space programs, opening doors for unprecedented scientific research, entrepreneurial ventures, and technological advancements (Ahmed Khan et al., 2022). Despite their diminutive frames, small satellites have yielded monumental achievements, propelling remarkable progress in navigation, remote sensing, Earth observation, telecommunications, climate monitoring, and many other fields of study (Yu et al., 2017).

CubeSats were initially devised as an educational tool at Stanford University and California Polytechnic State University, seeking to facilitate hands-on training for students in the field of space engineering. University-built small satellites represent an impressive convergence of academic excellence and technological innovation (Lopez-Salamanca et al., 2022). These satellite projects, developed within the academic environment, showcase the ingenuity and expertise of students and researchers. With ambitious goals and limited resources, university teams embark on the design, construction, and launch of small satellites that contribute to scientific research, technological advancements, and educational opportunities. These projects foster interdisciplinary collaboration, providing valuable hands-

on experience for students across various fields, including engineering, physics, computer science, and more. University-built small satellites demonstrate the power of academia in pushing the boundaries of space exploration and contributing to our understanding of the cosmos (Yu et al., 2022, 2018). These endeavors not only ignite a passion for space among the next generation of scientists and engineers but also strengthen university-industry partnerships and pave the way for future advancements in the field.

The utilization of Commercial Off-The-Shelf (COTS) components has revolutionized the development of small satellites (Dahbi et al., 2017; Guldager et al., 2005; O'Halloran et al., 2017; Selčan et al., 2017). These readily available, pre-existing components from the commercial market provide numerous benefits to small satellite missions. COTS components offer cost-effectiveness, as they eliminate the need for custom-designed hardware and reduce development lead times. By leveraging the advancements made in the consumer electronics industry, small satellite developers can access high-performance, miniaturized, and reliable components such as sensors, processors, power systems, and communication modules. This enables university teams, research institutions, and private enterprises to focus their efforts on mission-specific requirements and scientific objectives instead of reinventing the wheel

(Ahmed Khan et al., 2021). The use of COTS components has accelerated the pace of innovation in small satellite development, democratizing access to space and driving unprecedented progress in space exploration. While the use of COTS components has proven beneficial in small satellite development, there are inherent challenges related to their reliability and stability. Small satellites operate in harsh space environments, subjecting COTS components to extreme temperatures, radiation, and vacuum conditions. These off-the-shelf components are primarily designed for terrestrial applications and may not inherently possess the robustness required for the rigors of space. Concerns arise regarding their long-term reliability and stability in the space environment, including potential radiation-induced failures, sensitivity to temperature variations, and susceptibility to the vacuum of space. Ensuring the dependability of COTS components often involves extensive qualification testing, environmental hardening, and modifications to mitigate potential risks. Additionally, the limited availability and frequent obsolescence of certain COTS components pose challenges in mission longevity and spacecraft maintenance. Therefore, while COTS components offer cost and time advantages, careful consideration, thorough testing, and incorporating redundancy are essential to ensure their reliability and robustness in small satellite missions.

Small satellites are equipped with a range of actuators that play key roles in their operations. Reaction wheels are momentum exchange devices that allow for precise attitude control by spinning rapidly or slowing down to adjust the spacecraft's orientation (Jackulin Mahariba et al., 2022). Magnetic actuators, such as magnetorquer rods, interact with the Earth's magnetic field to generate torque, enabling attitude maneuvers and unloading accumulated momentum (Shoaib et al., 2022). Thrusters are propulsion systems that generate thrust to maneuver and control the spacecraft in space (Gill et al., 2013). Thrusters offer key advantages for small satellites, providing maneuverability, precise attitude control, and orbital adjustments. They enable critical functions like attitude maneuvers, orbit insertion, station-keeping, collision avoidance, and controlled reentry and debris mitigation. Thrusters operate with various propellants, including green options, aligning with sustainability goals. Japanese company Pale Blue recently conducted a successful test of its water vapor propulsion system, using it to fly a nano-satellite in space (Blue, 2023). The environmentally friendly propulsion system, tested on Sony's Star Sphere 1 satellite, aims to provide more affordable and sustainable options for small spacecraft. The system split water into hydrogen and oxygen, which were then burned as fuel, creating a cost-efficient and safe option for satellite propulsion. While the technology is best suited for small, low-cost satellites, it represents a significant step towards reinventing the small satellite industry. IoT solutions provider Hiber successfully launched their 3U CubeSat, Hiber Four, using a green propulsion system developed by Dawn Aerospace and

Hyperion Technologies. This innovative system allows Hiber Four to leverage cost-effective launches while maintaining operational flexibility. The propulsion system enables faster orbital adjustments and outperforms traditional systems in terms of performance, toxicity, and cost. Supported by the European Space Agency and Netherlands Space Office, Hiber Four showcases responsible satellite operations, addressing concerns regarding space debris and offering improved maneuverability and deorbiting capabilities. The green propellant, based on nitrous oxide and propene, delivers superior efficiency compared to alternative options while ensuring compliance with emerging global policies (Blue, 2023). Boswell Technologies and the Australian National University (ANU) successfully deployed their naphthalene-powered thruster into orbit as part of the payload on a Skykraft satellite launched by SpaceX (Dawn Aerospace and Hyperion Technologies., 2023). The innovative 'Bogong 1 Thruster' uses naphthalene, a safe and low-cost hydrocarbon, as a propellant. The transition of naphthalene from a solid to a gas produces thrust for satellite movement. While these technologies are paving the way for the small satellites industry, they are associated with serious risks that could jeopardize important missions as well. NASA's Lunar Flashlight spacecraft, designed to search for water ice deposits at the lunar south pole, will be unable to complete its science mission due to a problem with its propulsion system (NASA's Jet Propulsion Laboratory., 2023). The briefcase-sized spacecraft successfully demonstrated advanced technologies, such as a low-power flight computer and upgraded radio, which will benefit future missions. However, the spacecraft's novel propulsion system, consisting of four tiny thrusters, encountered issues in generating sufficient power for precise orbit insertion around the Moon. It was suspected that debris in the fuel lines hindered the consistent thrust generation. While attempts were made to optimize one thruster, the other three did not perform consistently despite similar procedures. This shows that the miniaturized thruster systems can come at cost which necessitates the need of stable and reliable attitude control system as to not risk the mission integrity. Furthermore, while advances are certainly been made in the field of miniaturized propulsion systems most university built CubeSats use reaction wheels combined with magnetorquers for the task of detumbling maneuvers.

Upon spacecraft deployment, residual angular momentum from the launch vehicle can result in an unstable state characterized by a high tumbling rate (Khan et al., 2023). To stabilize the spacecraft, magnetorquer rods are utilized. These rods generate opposing magnetic torque, gradually reducing the tumbling rate to an acceptable level. Once stability is achieved, precision and control authority are handed over to the reaction wheels, becoming the primary means of attitude control. At this stage, the magnetorquers are typically deactivated to prevent interference. However, continued operation of the reaction wheels can induce momentum accumulation, leading to saturation. In

response, magnetorquers are periodically reactivated in a desaturation process. By energizing the magnetorquer coils, additional control torque is provided to unload accumulated momentum, optimizing reaction wheel performance and maintaining stability. The desaturation process is instrumental in maintaining optimal operational conditions for the reaction wheels, resulting in improved attitude control stability. The research done by Anwar *et al.* and Mughal *et al.* presents a disruptive technique, replacing traditional 3D magnetic torquer rods with compact and lightweight embedded planar magnetorquer coils made from PCB copper traces (Ali *et al.*, 2015; Mughal *et al.*, 2020). These coils are constructed as printed circuit board (PCB) copper traces integrated within the internal layers of the PCB. This design offers a compact and lightweight alternative with a better torque-to-power dissipation ratio compared to traditional torquer rods. However, while these printed magnetorquers have advantages such as reduced mass and improved torque efficiency, their restricted actuation torques pose challenges for detumbling high initial angular rates in CubeSats. Ali *et al.* and Mughal *et al.* focused on printed magnetorquers with symmetric planar metal windings (Ali *et al.*, 2015; Mughal *et al.*, 2020). However, restricted actuation torques generated by symmetric topologies pose challenges for detumbling CubeSats with high initial angular rates. The work done in (Ahmed Khan *et al.*, 2023) used non-uniform geometric width winding distributions to detumble CubeSat Standard spacecraft but did not incorporate reaction wheels in their system design for optimal performance. To address this limitation, this work proposes combined use of reaction wheels and asymmetric printed magnetorquers for detumbling high form factor demanding multi-cube vehicles. Through this combined approach, the stability and control of a miniaturized spacecraft can be improved, ensuring their successful mission execution. In this paper, the selected technique for spacecraft detumbling is the well-established projection-based form of the standard B-dot control, as frequently reported in the literature (Lovera, 2015; Sanyal and Lee-Ho, 2009). This choice is advantageous as it reduces the computational demands placed on the onboard computer subsystem (OBC).

The schematic of system level design is illustrated in Fig. 1. The block diagram presented in Fig. 1(a) outlines the intercommunication between the Attitude Control System (ACS) and the Attitude Determination System (ADS). Variations in trace widths for Coils C1, C2, and C3 are implemented using a monomial constant, monotonically increasing and decreasing functions, respectively. Coils C4 and C5 follow the quadratic function with the width of the trace at maximum and minimum in the central turns, as depicted in Fig. 1(b)–(f). The cross sectional view visually depicted in Fig. 1(h) shows the magnetorquer copper windings internally embedded inside the inner layers of PCB.

No prior research has delved into the investigation of a reliable and fail-proof reconfigurable attitude control design of reaction wheels alongside embedded asymmetric

planar windings, specifically for CubeSats, a void this paper aims to fill. With around 50 % of ACS failures being linked to faults in reaction wheel moving parts, there is a deliberate focus on developing strategies to tackle potential worst-case reaction wheel failures (Jin *et al.*, 2012).

The structure of the paper is outlined as follows: The analytical model of the magnetorquer and reaction wheels is presented in Section 3, while the equations of motion pertaining to attitude are also presented. The detumbling maneuvers of small satellites, carried out using a combined scheme consisting of reaction wheels and diverse planar magnetorquers with nonuniform geometric metal width distribution, are elaborated in Section 4. Additionally, a comprehensive thermal analysis of the magnetorquers is performed, focusing on validating the safety of heat transfer and conducting a comparative analysis with the best-case designs.

1.1. Contributions

- The primary contribution is the design and development of a novel ACS integrated into a CubeSat-compatible PCB, leveraging COTS reaction wheels and asymmetric magnetorquer planar windings.
- This research proposes and explores various detumbling schemes that exploit the reconfigurable nature of embedded asymmetric magnetorquers in hybrid combination with reaction wheels, enhancing the system's adaptability and fault tolerance.
- A key contribution is the demonstration of the asymmetric embedded magnetorquer's ability to effectively detumble the spacecraft in the event of reaction wheel faults, such as bearing issues, imbalance, wear, or motor failures, without relying on auxiliary compensators.
- The designed PCB features a modular, redundant, radiation-hardened, and simple design, ensuring high reliability and performance in space environments.

2. Mathematical model

The mathematical model used here utilizes the Earth Magnetic Field and Gravitational Models (Geographic library c++, 2019). The magnetic field model utilized is derived from the well-established international geomagnetic reference field (IGRF) model. Similarly, the gravitational model employed is based on the widely accepted EGM2008 model. Quaternions are employed to parameterize the orientation of the satellite, assuming it behaves as a rigid body. Reaction wheels and Magnetorquers are modeled to test controllers assuming a certain amount of current and number of turns in the coils.

2.1. Reaction wheel model

The direct influence of the reaction wheels on the vehicle's inertia mandates the incorporation of the reaction

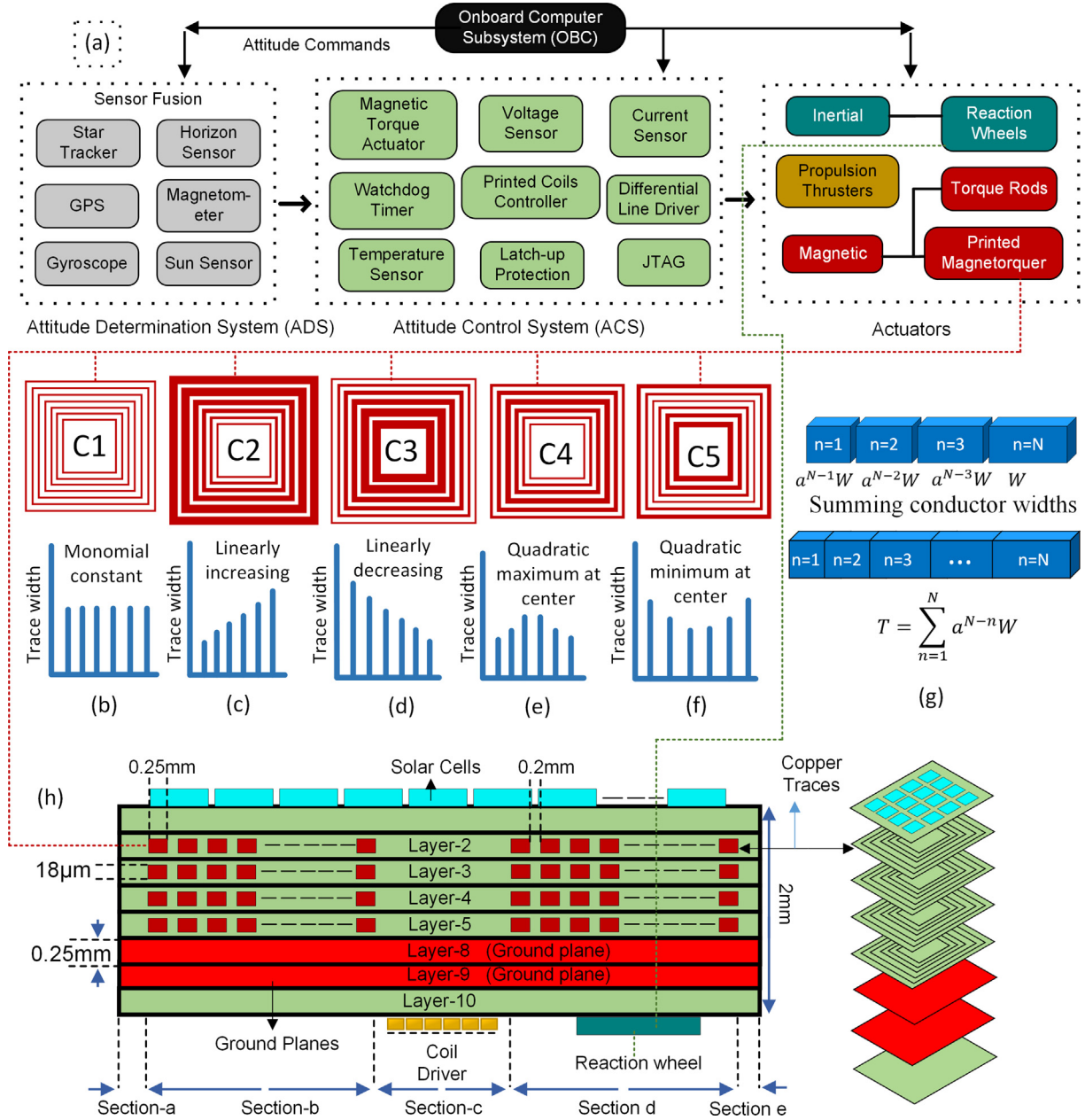


Fig. 1. (a) The intercommunication between the ACS and ADS is facilitated through the OBC. (b)–(f) Magnetorquer windings layouts with five arrangements based on mathematical functions. (g) TWR (Total Width Ratio) is utilized to evaluate the cross section of trace widths. In this context, T represents the cumulative width of traces within a single layer, which remains consistent for a specific footprint and number of turns. (h) Cross-sectional view of printed magnetorquer PCB having reaction wheel.

wheel model ahead of analyzing the attitude dynamics. The assumed vehicle incorporates three reaction wheels, with each one exhibiting its distinctive angular velocity ω_{Ri} and angular acceleration α_{Ri} . The Eq. (1) provided below defines the inertia of each reaction wheel relative to its center of mass, assuming the reaction wheel to be a disk with a specific radius r_{RW} and height h_{RW} . The superscript R is used to represent the frame of reference, while the subscript R shows that the inertia matrix is about the center of mass.

$$I_{Ri}^R = \begin{bmatrix} m_R r^2 / 2 & 0 & 0 \\ 0 & (m_R / 12)(3r_{RW}^2 + h_{RW}^2) & 0 \\ 0 & 0 & (m_R / 12)(3r_{RW}^2 + h_{RW}^2) \end{bmatrix} \quad (1)$$

An axis of reaction wheel rotation is used for rotating the inertia of matrix into the vehicle body frame of reference. The vector $\hat{n}_{Ri}$ shows the axis about which the reaction wheel is supposed to rotate. Euler angles θ_{ri} and ψ_{ri} can be extracted from this unit vector. The rotation matrix

$\mathbf{T}_{Ri}(0, \theta_{ri}, \psi_{ri})$ is then generated using the transformation matrix from the body to inertial called 3-2-1 sequence given in Eq. (2). The shorthand notations are used for trigonometric functions such as: $\cos(\alpha) \equiv c_\alpha$, $\sin(\alpha) \equiv s_\alpha$, and $\tan(\alpha) \equiv t_\alpha$. The matrix is then used to calculate the reaction wheel inertia in the vehicle body frame.

$$\mathbf{T}_{IB}(\phi, \theta, \psi) = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix} \quad (2)$$

$$\mathbf{I}_{Ri}^B = \mathbf{T}_{Ri}^T \mathbf{I}_{Ri}^R \mathbf{T}_{Ri} \quad (3)$$

The parallel axis theorem is then utilized for shifting the inertia to the spacecraft's center of mass where the subscript RB represents the inertia of the reaction wheel inertia taken about the spacecraft's vehicles center of mass.

$$\mathbf{I}_{RBi}^B = \mathbf{I}_{Ri}^B + m_{Ri} \mathbf{S}(\vec{r}_{Ri}) \mathbf{S}(\vec{r}_{Ri})^T \quad (4)$$

The vector $\vec{r}_{Ri}$ is the distance from the center of mass of the spacecraft to the center of mass of the reaction wheel in the vehicle body reference frame. The total inertia of the reaction wheels and spacecraft is then simply a sum of all the reaction wheel inertia.

$$\mathbf{I}_S = \mathbf{I}_B + \sum_{i=1}^3 \mathbf{I}_{RBi}^B \quad (5)$$

The total angular momentum of the vehicle is equal to the equation mentioned below where $\vec{\omega}_{B/I}$ is the angular velocity of the spacecraft.

$$\vec{H}_S = \mathbf{I}_B \vec{\omega}_{B/I} + \sum_{i=1}^3 \mathbf{I}_{Ri}^B \omega_{Ri} \hat{n}_{Ri} \quad (6)$$

$$\mathbf{T}(\vec{q}) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 + q_0 q_3) & 2(q_1 q_3 - q_0 q_2) & 2(q_1 q_2 - q_0 q_3) q_0^2 - q_1^2 + q_2^2 - q_3^2 \\ 2(q_0 q_2 + q_1 q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_0 q_3 - q_1 q_2) & 2(q_0 q_3 + q_1 q_2) q_0^2 - q_1^2 - q_2^2 + q_3^2 \\ 2(q_1 q_3 - q_0 q_2) & 2(q_0 q_3 - q_1 q_2) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_0 q_2 + q_1 q_3) \\ 2(q_0 q_2 + q_1 q_3) & 2(q_0 q_3 + q_1 q_2) & 2(q_0 q_2 - q_1 q_3) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix} \quad (11)$$

Using a similar methodology, the total torque applied to the vehicle can be derived based on the following equation.

$$\vec{M}_R = \sum_{i=1}^3 \mathbf{I}_{Ri}^B \alpha_{Ri} \hat{n}_{Ri} \quad (7)$$

It is widely accepted that the angular acceleration of each reaction wheel can be controlled directly, yet as the angular velocity of the reaction wheel rises, the maximum attainable angular acceleration progressively diminishes. Once the reaction wheel reaches its maximum angular velocity, the angular acceleration gradually reduces to zero, a state referred to as reaction wheel saturation. To address this, a technique called momentum dumping is employed. However, the desaturation scheme, which tackles this issue, is not covered in this paper.

2.2. Asymmetric magnetorquer model

Achieving spacecraft control using the asymmetric planar coils demands the utilization of three sets of coils, arranged in an orthogonal orientation. The alignment of each coil within this arrangement with the principal axes of the spacecraft's body frame offers accurate control and

alignment capabilities. By independently controlling each planar asymmetric coil through the application of respective currents $\vec{i}_M = [i_x, i_y, i_z]^T$, the desired magnetic moment can be generated. The magnetic moment is represented by the following expression;

$$\vec{\mu}_M = nA \vec{i}_M \quad (8)$$

where the value of n signifies the number of turns incorporated in the coil of each magnetorquer, while A denotes the area. The torque produced by the magnetorquers in the Body reference frame is expressed as the vector cross product between Earth's magnetic field and the magnetic dipole moment.

$$\vec{M}_M = \mathbf{S}(\vec{\mu}_M) \mathbf{T}(\vec{q}) \vec{B}_I \quad (9)$$

The quaternion transformation matrix, as expressed in the Eq. (10) to convert the inertial magnetic field vector into the spacecraft's body frame. This transformation facilitates the determination of the magnetic field vector within the spacecraft's reference frame. The transformation matrix is calculated as;

$$\begin{Bmatrix} \hat{q}_0 \\ \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{Bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix} \begin{Bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{Bmatrix} \quad (10)$$

where four quaternions are represented by q_i and p, q, r denote the angular velocities in the body frame of the vehicle. The quaternions are subsequently expressed in the form of a scalar (q_0) and a vector with three elements (q_1, q_2, q_3).

The application of the identity $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$ allows for a further simplification of the component form of Eq. (9) such that

$$\begin{Bmatrix} L \\ M \\ N \end{Bmatrix} = nA \begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix} \begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} \quad (12)$$

The body frame of the vehicle is associated with the magnetic field components B_x, B_y, B_z . The study (Khan et al., 2023) examines the magnetic dipole moments and the corresponding power dissipation resulting from different coil configurations. The magnetic dipole moment $\vec{D}$ of M_x magnetorquer coils configured in any desired reconfigurable topological arrangement (n-series, m-parallel, $n \times m$ -hybrid) is given by

$$\vec{D} = M_x \vec{D}_0 \quad (13)$$

where $\vec{D}_0$ represents magnetic moment of a single coil. Similarly, single coil's power consumption is then given by P_0

$$\therefore P_0 = I_0^2 R_0, R_0 = \rho \frac{L_t}{A}$$

Power dissipation of M_x coils arranged in any desired arrangement is (n-series, m-parallel, $n \times m$ -hybrid) is;

$$P = M_x P_0 \quad (14)$$

$$\therefore I = m I_0, V = n V_0, M_x = n \times m$$

The ratio between magnetic dipole moment generated and power dissipated is represented by the following equations;

$$\frac{D_0}{P_0} = \frac{N.S.I_0}{R_0.I_0^2} = \left(\frac{N.S}{R_0} \right) \cdot \frac{I}{I_0} \quad (15)$$

$$D_0^2 = \frac{(N.S)^2}{R_0} \cdot P_0 \quad (16)$$

$$D^2 = n.m. \frac{(N.S)^2}{R_0} \cdot P = c.P \quad (17)$$

The parameter c or $\frac{D^2}{P_0}$ serves as a performance evaluation metric that relies on the number of coils and their dimensions when connected in series or parallel configurations. Increasing the number of coils, regardless of the configuration (parallel, series, or hybrid), results in higher power dissipation and magnetic dipole moment. The current passing through the magnetorquers remains constant in single or series configurations, generating similar levels of torque.

2.3. Attitude equations of motion

The attitude equations of motion are formulated with the assumption of the vehicle rotating about the three axes. The angular velocity's derivative is computed by equating the derivative of angular momentum to the total moments exerted on the spacecraft while the reaction wheel torques are added.

$$\dot{\vec{\omega}}_{B/I} = I_S^{-1} \left(\vec{M}_P + \vec{M}_R + \vec{M}_M - \mathbf{S}(\vec{\omega}_{B/I}) \vec{H}_S - \dot{I}_s \vec{\omega}_{B/I} \right) \quad (18)$$

The term I_s is the change in inertia in the body frame caused by deployment of solar panels and/or antenna. The moments applied use the subscripts P for propulsion, M for magnetorquers, and R for reaction wheels. The total angular momentum, denoted by $\vec{H}_S$, incorporates the combined effect of the reaction wheels in the overall vehicle dynamics. Where $\mathbf{S}$ is the skew symmetric operator and multiplying it with a vector results in a cross product

$$\mathbf{S}(\vec{\omega}_{B/I}) = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix} \quad (19)$$

2.4. B-Dot controller

The B-dot controller is an attitude control technique used in satellites. It utilizes the time derivative (B-dot) of the measured magnetic field to generate control signals for adjusting the satellite's attitude. By monitoring changes in the magnetic field, the B-dot controller can detect and counteract unwanted rotations, improving the stability and orientation of the satellite. The implementation of the projection-based B-dot controller involves utilizing a simple control law for comparing the detumbling rates and behavior of small satellites equipped with PCB-integrated reconfigurable coils, as demonstrated in Eq. (20) (Monkell et al., 2018).

$$\vec{\mu}_B = K \mathbf{S}(\vec{\omega}_{B/I}) \mathbf{T}(\vec{q}) \vec{B}_I \quad (20)$$

The controller involves projecting or crossing the angular velocity vector $\vec{\omega}_{B/I}$ from the inertial frame to the body frame onto the magnetic field in the body frame using the quaternion transformation matrix $\mathbf{T}_{(q)} \rightarrow$. The result is then

scaled by the factor K to calculate the magnetic dipole moment $\vec{\mu}_B$ in the body frame. Eq. (12) provides the representation of the currents in the components form, incorporating the application of the identity $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$.

$$\begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} = \frac{K}{NA} \begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix} \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \quad (21)$$

$$\begin{Bmatrix} L \\ M \\ N \end{Bmatrix} = -K \begin{bmatrix} B_y^2 + B_z^2 & -B_x B_y & -B_x B_z \\ -B_x B_y & B_x^2 + B_z^2 & -B_y B_z \\ -B_x B_z & -B_y B_z & B_x^2 + B_y^2 \end{bmatrix} \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \quad (22)$$

Spacecraft's total torque is computed by substituting Eq. (21) into Eq. (12) under the assumption that the planar coils could generate the required current.

The objective of the controller is to minimize the angular velocities and bring them to a near-zero state, however, the literature indicates that this outcome is not fully achieved. Several explanations account for this observation. Firstly, Eq. (9) assumes that the magnetic moment is not aligned with the Earth's magnetic field. If the magnetic moment is collinear with the magnetic field, the result is a zero-torque applied to the satellite. Moreover, if the angular velocity vector of the satellite aligns with the magnetic field, Eq. (21) yields zero current. However, during the detumbling state, it is crucial to consider that the Earth's magnetic field is time-varying and continuously changing as a result of the satellite's orientations or orbital changes experienced. around all axes. Based on this analysis, it can be inferred that the effectiveness of the B-Dot control law in stabilizing the spacecraft's angular momentum relies on the non-parallel alignment of the angular velocity vector $\vec{\omega}_B$ and the Earth's magnetic field vector $\vec{B}_I$.

2.5. Reaction wheels control

Under the assumption that each reaction wheel is aligned with a principal axis of inertia, the control scheme is characterized by its simplicity. In the analysis, we proceed with the assumption that the reaction wheel can be subjected to torque, enabling direct control over its angular velocity α_{Ri} . Given that the satellites align with the principle axes of rotation, a basic PD control law is implemented to facilitate the spacecraft's orientation to various desired orientations, incorporating Euler angles within the control law.

$$\alpha_{Ri} = -k_p(\epsilon_i - \epsilon_{desired}) - k_d(\omega_i - \omega_{desired}) \quad (23)$$

The value of ϵ in the equation depends on the reaction wheel being referenced and can correspond to the roll angle ϕ , the pitch angle θ , or the yaw angle ψ . To calculate the Euler angles, it is required to convert the quaternions into the Euler angle representation, enabling integration within the equation mentioned above.

$$\phi = \tan^{-1} \left(\frac{2(q_0 q_1 + q_2 q_3)}{1 - 2(q_1^2 + q_2^2)} \right) \quad (24)$$

$$\theta = \sin^{-1}(2(q_0 q_2 - q_3 q_1)) \quad (25)$$

$$\psi = \tan^{-1} \left(\frac{2(q_0 q_3 + q_1 q_2)}{1 - 2(q_2^2 + q_3^2)} \right) \quad (26)$$

In many cases, the orientation of reaction wheels may not align precisely with the principal axis of inertia. In such cases, the implementation of a Least Squares Regression model becomes necessary. Therefore, Eq. (23) plays a crucial role in calculating the desired torque to be placed on the satellite, ensuring efficient and accurate maneuvering in space.

$$\vec{M}_{desired} = -k_p(\epsilon_i - \epsilon_{desired}) - k_d(\omega_i - \omega_{desired}) \quad (27)$$

By simplifying the expression, we derive the equation for the total torque applied to the spacecraft, incorporating the combination of angular accelerations into a vector format.

$$\vec{M}_{desired} = \vec{M}_R = \sum_{i=1}^{NR} \mathbf{I}_{Ri}^B \alpha_{Ri} \hat{n}_{Ri} = [\mathbf{I}_{Ri}^B \hat{n}_{R1} \cdots \mathbf{I}_{RNR}^B \hat{n}_{RNR}] \begin{Bmatrix} \alpha_1 \\ \vdots \\ \alpha_{NR} \end{Bmatrix} \quad (28)$$

$$\vec{M}_{desired} = \vec{M}_R = [\mathbf{I}_{Ri}^B \hat{n}_{R1} \cdots \mathbf{I}_{RNR}^B \hat{n}_{RNR}] \begin{Bmatrix} \alpha_1 \\ \vdots \\ \alpha_{NR} \end{Bmatrix} = \mathbf{J} \vec{\alpha} \quad (29)$$

The impossibility of directly inverting matrix $\mathbf{J}$, with its dimensions of $3 \times N_{RW}$, prevents a straightforward solution for the vector of angular accelerations $\vec{\alpha}$. Hence, in order to find the solution with the least amount of angular accel-

eration, a minimization routine is necessary. LaGrange's method is employed in this case to determine the vector of angular acceleration.

$$\vec{\alpha} = \mathbf{J}^T (\mathbf{J} \mathbf{J}^T)^{-1} \vec{M}_{desired} \quad (30)$$

2.6. Temperature profile

The calculation of dissipated power and the resulting temperature rise for the PCB-integrated reconfigurable embedded planar coils is performed based on the mathematical framework derived in reference (Ali et al., 2015; Mughal et al., 2020), which is represented as follows: The heat balance equation is utilized in the examination of a stable system, allowing for the correlation between the effective temperature resulting from the external environment $T_{external}$ and the temperature rise caused by internal heat dissipating elements, including the planar coils $T_{internal}$. The total thermal heat experienced by the CubeSat due to the external environment, $T_{external}$, can be determined by employing Eq. (31).

$$T_{external}^4 = \frac{q_s}{\sigma} \left(\frac{1}{4} + F\alpha \right) + FT_E^4 \quad (31)$$

The radiation heat transfer in the PCB is governed by Stefan Boltzmann's Law, which can be represented by equation (30).

$$T_{internal}^4 = \frac{P}{\sigma \epsilon A} \quad (32)$$

The computation of the CubeSat's overall temperature can be achieved by utilizing Eqs. (31) and (32).

$$T^4 = T_{external}^4 + T_{internal}^4 \quad (33)$$

$$T = \sqrt[4]{\frac{q_s}{\sigma} \left(\frac{1}{4} + F\alpha \right) + FT_E^4 + \frac{P}{\sigma \epsilon A}} \quad (34)$$

where

$$q_s = 1371 \text{ W/m}^2, \sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4, F = 0.15, \text{ and } T_E = 255 \text{ K}$$

The analysis quantifies the temperature rise resulting from the asymmetric planar coils, with the exclusion of any other internal heat dissipations.

3. System design description

The integration of the planar magnetorquers is accomplished by seamlessly fitting them into the four inner layers of the eight-layered solar panel PCB. This integration approach does not result in any additional space requirements on the spacecraft. A visual depiction of this integration can be seen in Fig. 2(a). The embedded coils within the PCB coil driver prototype, as depicted in Fig. 2(b), are subjected to varied voltage levels ranging from 1 V to 12 V. These voltage levels closely replicate the typical range used in power distribution buses found in CubeSat platforms. In Fig. 2(c), the voltages needed for various configurations of coils that provide constant current are displayed. Fabri-

cated planar magnetorquer PCBs with scaling factor of 0.95 applied to various geometric width distributions of the windings are presented in Fig. 2(d). Fig. 2(e) provides a cross-sectional view of the ACS module, offering insight into the arrangement of the reaction wheels. It is worth mentioning that these reaction wheels are actuated using the same driver used for the planar magnetorquers. The coil actuator is connected to a reconfigurable circuit, employing MOSFET switches, to facilitate design reconfigurability (1×4 , 4×1 , 2×2), as shown in the coil driver schematic in Fig. 2(f). The onboard processor allows for

configuration changes (1×4 , 4×1 , 2×2), granting the design better flexibility. Consequently, the OBC and telemetry processor unit (TPU) transmit telemetry data points to the ground station, tailoring the transmission to meet the mission's prerequisites regarding power consumption, torque, and coil-generated dipole moment.

The study conducted by the authors in (Ahmed Khan et al., 2023) focuses on investigating various geometric distributions of asymmetric planar magnetorquer coils with the objective of optimizing these configurations for CubeSats. The objective is to maximize the torque-to-power dis-

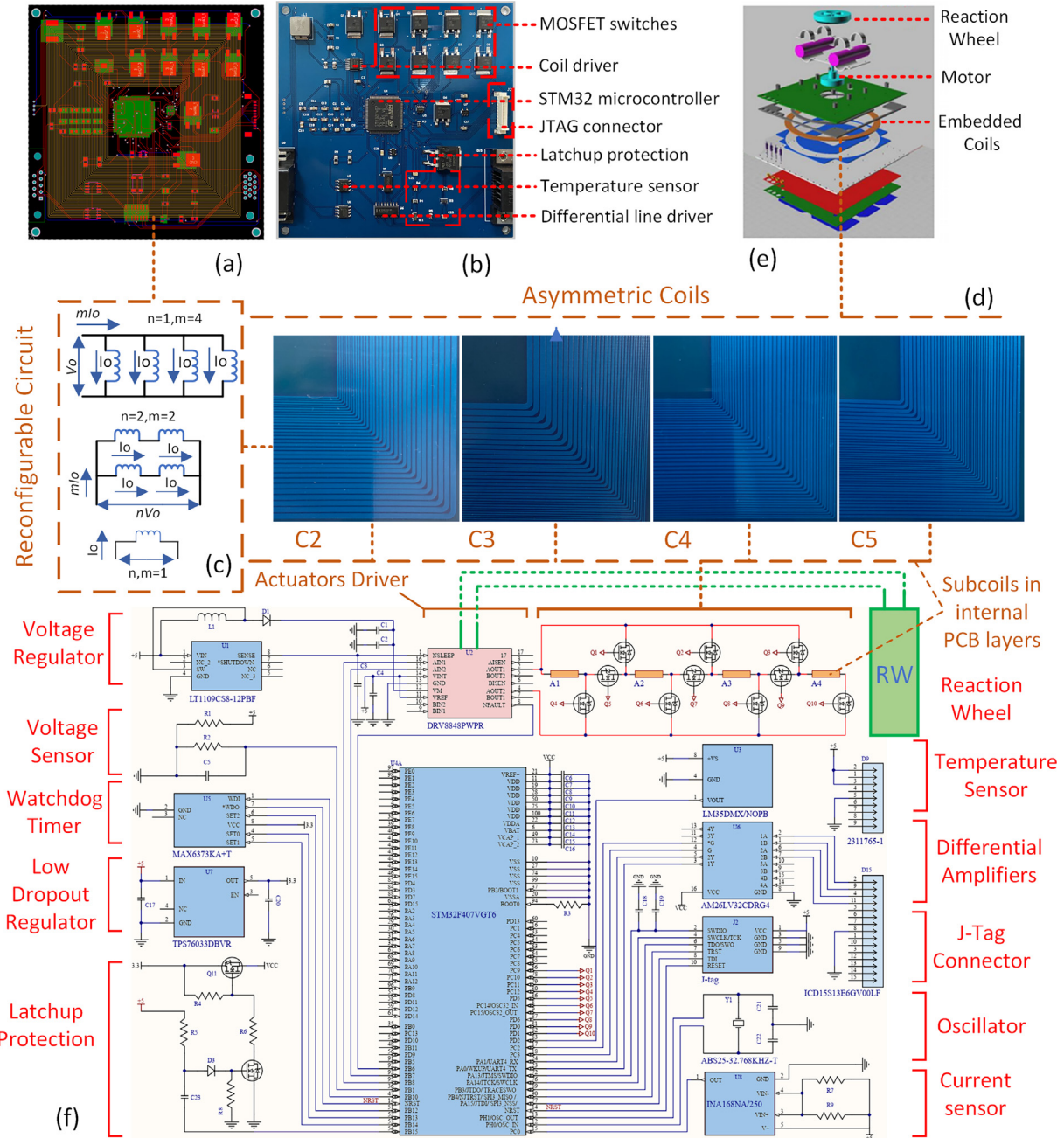


Fig. 2. (a) Planar coils embedded inside the multilayer PCB. (b) Coil driver module mounted on the bottom side of PCB (c) CubeSat illustration with embedded coils. (d) Voltages required for various configurations of coils that offer constant current. (e) Fabricated planar magnetorquer PCBs with TWR $a = 0.95$ applied to monomial constant C1, monotonously increasing C2, monotonously decreasing C3, quadratic maximum C4 and quadratic minimum C5 at central turns (f) Actuators driver schematic.

sipation ratio. Five asymmetric coil topologies with different widths are tested, occupying equal space on the PCB tile. They analyzed magnetic flux density, surface density losses, currents, and thermal gradients. Magnetorquer windings are modeled as boundaries using the Electric Currents in Shells interface using COMSOL Multiphysics software. The copper trace width varies based on mathematical functions (monomial constant, linearly increasing and linearly decreasing functions), while maintaining identical spacing between metal widths. The researchers employed the inner layers of a multi-layer PCB to integrate the magnetorquers, aiming to enhance the torque and magnetic moment (Ali et al., 2018, 2015). By implementing a scaling factor across different asymmetric configurations, it becomes possible to achieve significantly higher torques within a single layer, eliminating the need for additional internal PCB layers. The monotonously increasing C2 and decreasing C3 coils exhibit slight differences in terms of current drawn, resistance, power dissipation, magnetic energy, and surface electric potential. The quadratic maximum and minimum coils demonstrate nearly identical performance parameters when trace width values vary. The optimal geometric structure for a magnetorquer depends on the specific application and the balance between power density, efficiency, mission lifespan, and nanosatellite dimensions. Lower TWR (Trace Width Ratio) values generate significant torque at the expense of higher temperatures, potentially reducing the magnetorquer module's lifespan. Therefore, the selection of TWR values should consider the desired power ratings and mission lifespan. For missions with shorter lifespans and a need for exceptional pointing accuracy and faster rotational times, lower TWR values are advisable. Conversely, magnetorquer configurations with narrower winding traces, closer to unity TWR values, exhibit higher inductance, generate less heat, and are better suited for longer-span missions. The calculation of winding resistance for asymmetric geometric distributions with varying widths is performed using the methods outlined in reference (Huang et al., 1995).

4. Simulation results

In this section, the performance of the proposed system of asymmetric actuators and reaction wheels is evaluated through simulations conducted with two CubeSats. The first CubeSat is designed with a 3U form factor, measuring $10 \times 10 \times 30 \text{ cm}^3$ and weighing 3.2 kg, while the second CubeSat is configured as 6U, measuring $10 \times 10 \times 60 \text{ cm}^3$ and weighing 7.2 kg. The inertia values are computed as $I_x = 0.006 \text{ kgm}^2$, $I_y = 0.03 \text{ kgm}^2$, $I_z = 0.03 \text{ kgm}^2$ for the 3U CubeSat, and $I_x = 0.012 \text{ kgm}^2$, $I_y = 0.22 \text{ kgm}^2$, $I_z = 0.22 \text{ kgm}^2$ for the 6U CubeSat. The simulations are executed using a satellite orbiting at an altitude of 600 km with a 56° inclination. The simulation commences with an initial detumbling rate of $[0.5, -0.9, 0.9] \text{ rad/s}$,

which corresponds to a rotation speed of $51^\circ/\text{sec}$. The controller gain is determined by calculating the values using the design parameters of the coils, as outlined in Eq. (20). The Number of turns in single layer are 33 and 42 for monotonously increasing C2, and quadratic maximum C5, respectively. The current drawn of single coil is 0.023A, parallel arrangement (1×4) is 0.082A and the series arrangement (4×1) is 0.023A. The total current of planar magnetorquers cannot exceed these ratings. The authors recommend to revisit ref (Ali et al., 2015; Mughal et al., 2020) for more details about the performance parameters of the coils. The simulation employs a fixed step Runge-Kutta-4 integration routine with a time-step of 0.1, which ensures precise and reliable calculations, capturing the intricacies of the system dynamics. The assumption has been made that the magnetorquer coils allow for current to flow in both directions, enabling the spacecraft to perform three-dimensional attitude maneuvers. The reaction wheels activate when the angular velocities reach a safe value of lower than 0.5 rad/s . The shorthand notations are used for actuators such as magnetorquers $\equiv$ Mgs, reaction wheels $\equiv$ RWs in Figs. 3–9.

The reaction wheel utilized in this study has a maximum momentum of 0.015 Nms, a maximum torque of 4 mNm, a mass of 0.13 kg, dimensions of $42 \times 42 \times 19 \text{ mm}$, operates on a 10 V supply voltage, and has a peak power consumption of 5.5 W. In case of monotonously increasing Coil C2, PCB temperature is 38.5°C , imparted torque is $5.1 \mu\text{Nm}$, dipole moment generation is 0.10 Am^2 , power dissipated is 1.26 W, and the current is 0.37A. In monotonously decreasing C3 Coil's case, PCB temperature is 37.6°C , imparted torque is $4.8 \mu\text{Nm}$, dipole moment generation is 0.09 Am^2 , power dissipated is 1.18 W and the current is 0.35A. Quadratic maximum C4 and minimum C5 at central turns almost shows the same behavior. In case of quadratic maximum Coil C4, power dissipated is 0.8 W, PCB temperature is 34.14°C , the current is 0.34A, and torque exerted is $4.45 \mu\text{Nm}$, and generated dipole moment is 0.08 Am^2 . The torque rods produce a torque in the range of $3.0 \times 10^6 \text{ Nm}$, based on 84 turns, an area of 0.02 m^2 , and a current of 0.04 A. Figs. 3–6 and 7–9 reveals the detumbling strategies employed by the 3U and 6U CubeSats, respectively.

For asymmetric planar coils, changing the trace width affects the performance. A decrease in the TWR values leads to an increase in trace width, a decrease in the number of turns, and, subsequently, a reduction in resistance. This lowered resistance allows higher currents to flow through the coil windings, resulting in elevated temperatures. The C2 coil, which experiences a linear increase, draws the maximum current and generates greater torque, leading to faster satellite detumbling but with higher power consumption. On the other hand, the C5 coil, with a quadratic geometric distribution, draws less current and detumbles the satellite more slowly, resulting in lower power consumption.

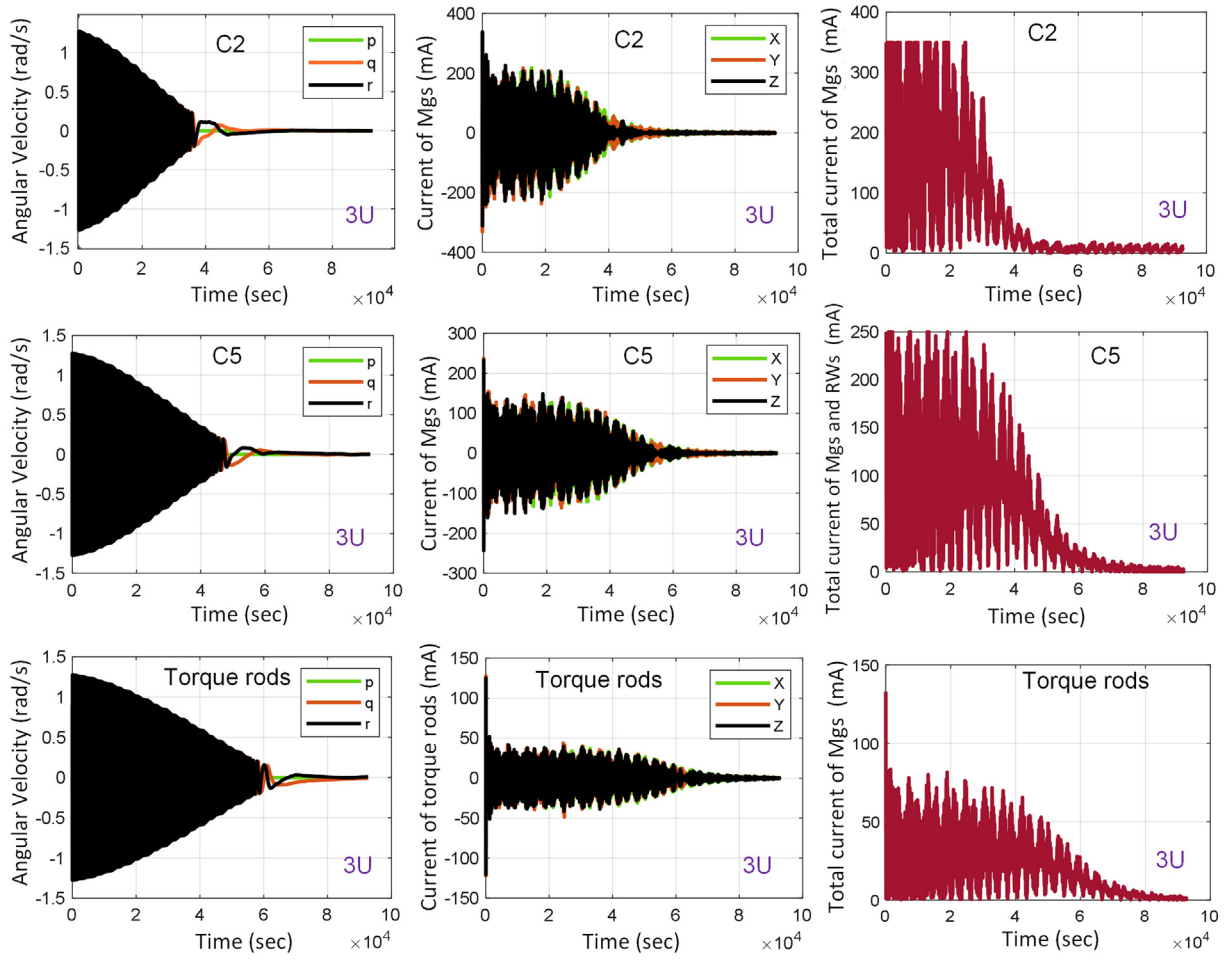


Fig. 3. Angular rates and current drawn in pitch, yaw and roll axes using only embedded Magnetorquers and torque rods.

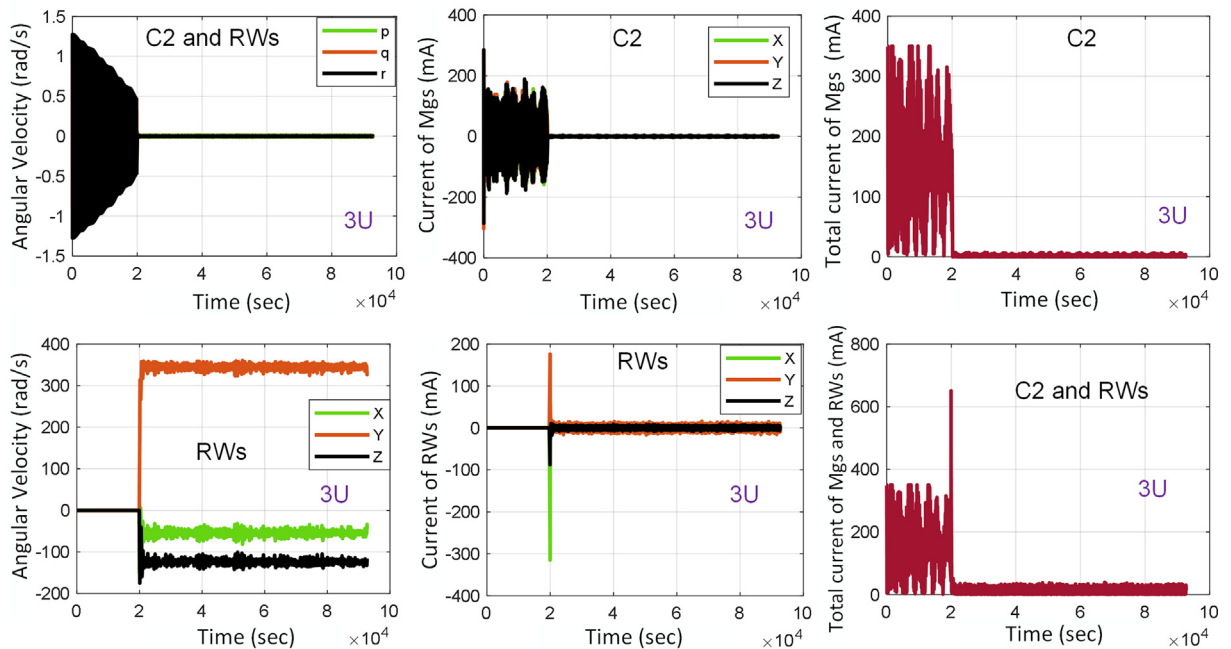


Fig. 4. Angular rates and current drawn in pitch, yaw and roll axes using embedded Magnetorquers (C2) with reaction wheels.

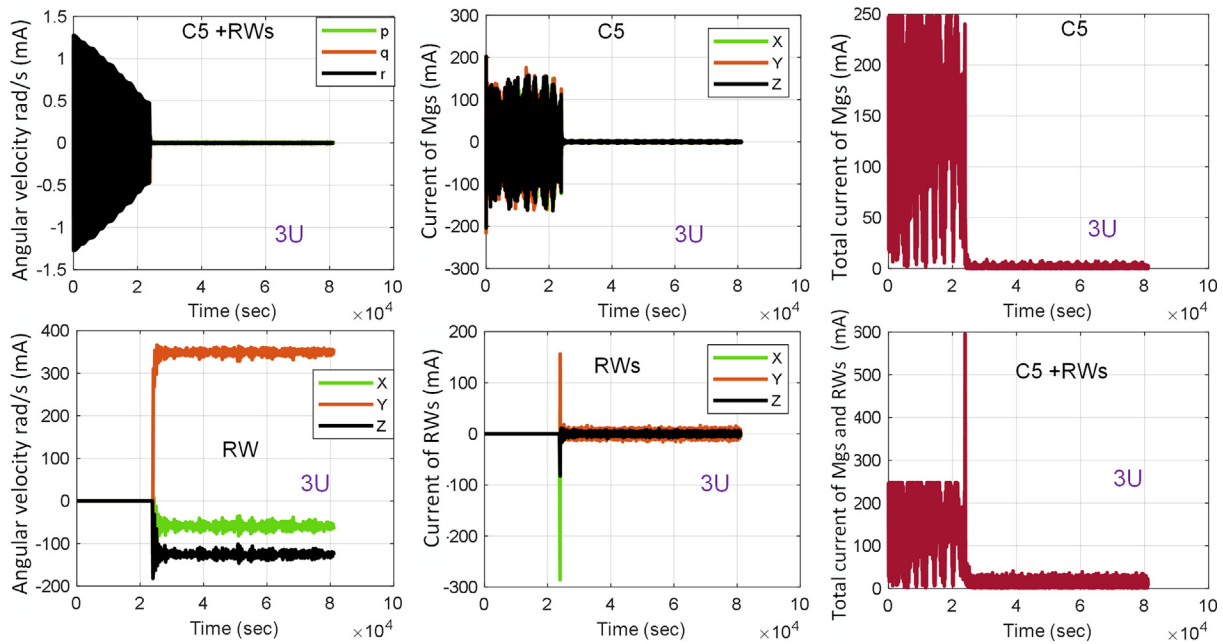


Fig. 5. Angular rates and current drawn in pitch, yaw and roll axes using embedded Magnetorquers (C5) with reaction wheels.

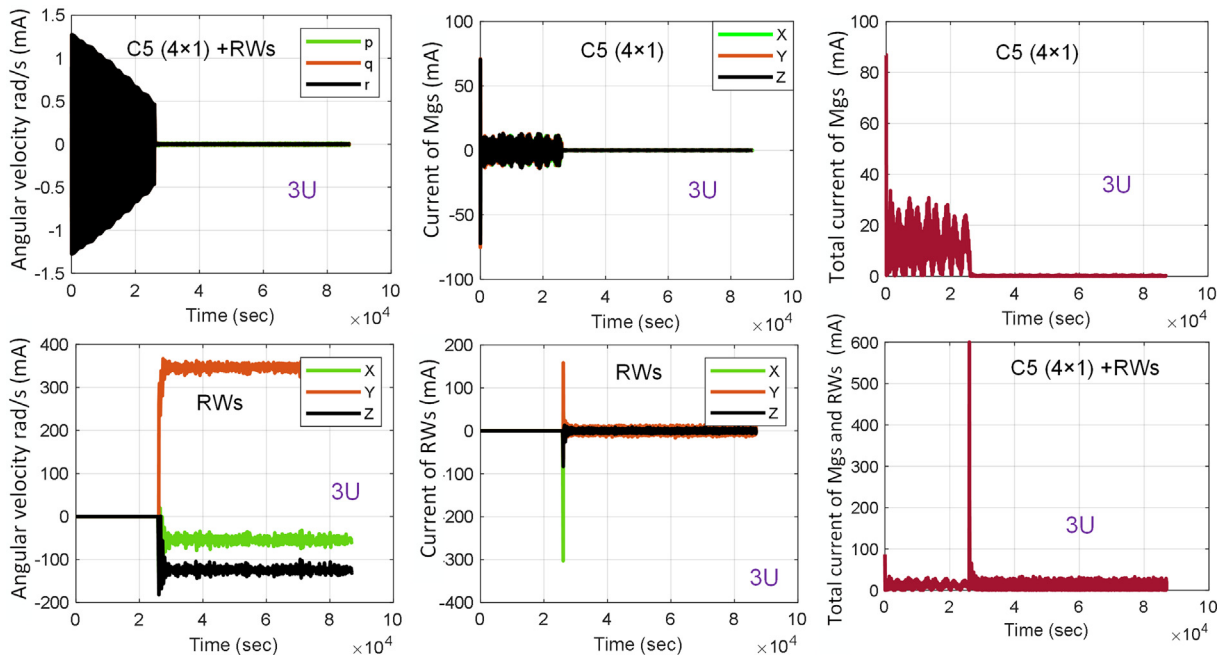


Fig. 6. Angular rates and current drawn in pitch, yaw and roll axes using Coil C5 in series (4×1) and reaction wheels.

Fig. 3 highlights the magnetorquers only scheme without using the reaction wheels for the task of detumbling. Although the planar coils detumble the vehicle significantly better than the torque rods, both the actuators take significant time to stabilize the angular rates. Hence, it is evident that the extended tumbling of high form factor spacecraft can pose a critical risk to the structural integrity of the subsystems and sensitive payloads. It becomes essential to address and resolve the detumbling process efficiently to mitigate potential damage and ensure the successful opera-

tion of the spacecraft. Therefore, reaction wheels are important in addressing the issue for abrupt task of detumbling the spacecraft.

Figs. 4–6 highlights the combined scheme of reaction wheels along with asymmetric coils. The initial phase of the detumbling is done through magnetorquers until they are saturated bringing norm to the point during which the reaction wheels are turned on (as soon as the combined angular velocity reaches a safe value of below 0.5 rad/s) and reduce the angular velocities to acceptable rates.

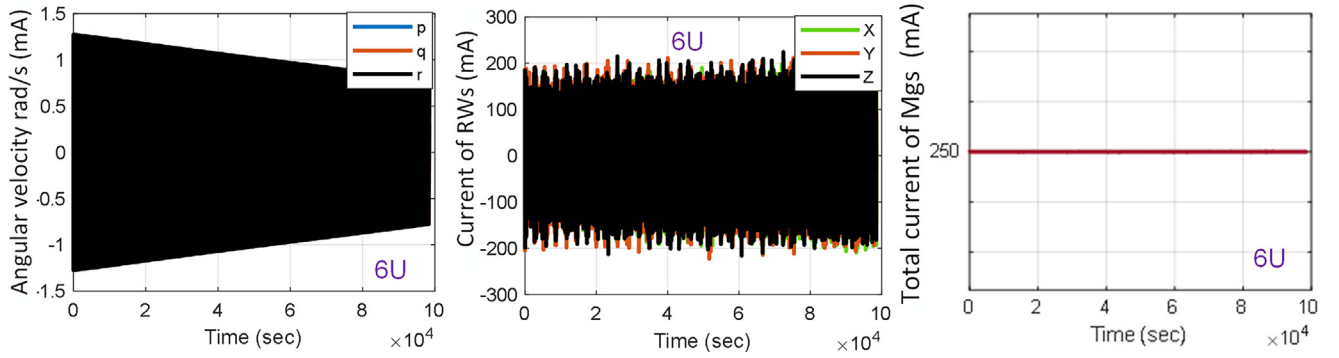


Fig. 7. Angular rates and current drawn in pitch, yaw and roll axes using Coil C5 for a 6U CubeSat (1×1).

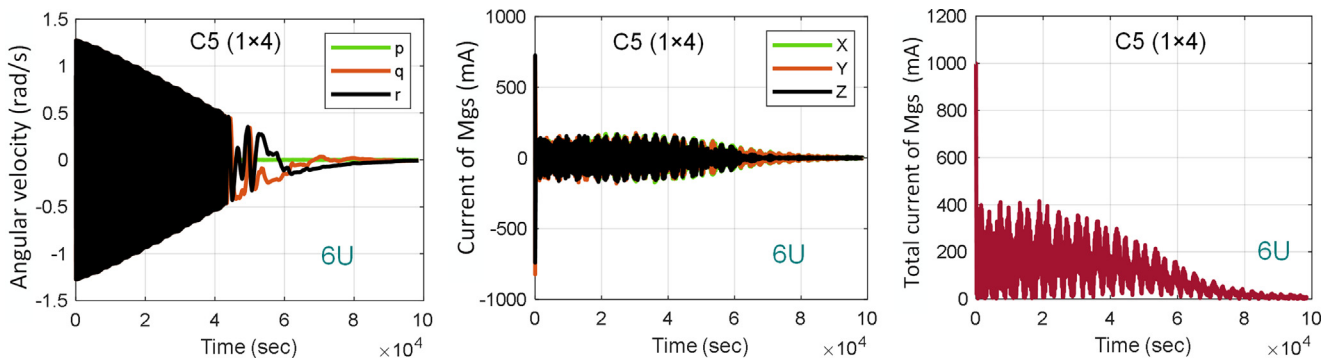


Fig. 8. Angular rates and current drawn in pitch, yaw and roll axes using embedded Magnetorquers C5 in parallel (1×4).

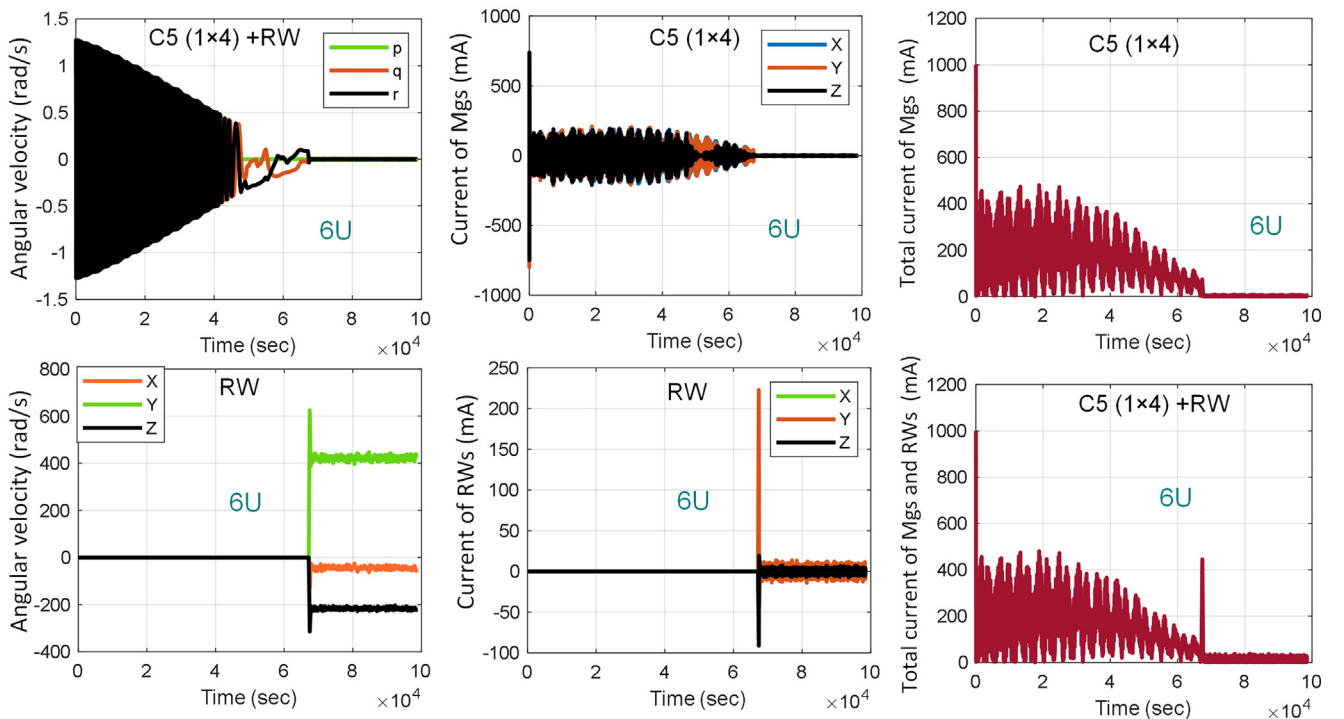


Fig. 9. Angular rates and current drawn in pitch, yaw and roll axes using embedded Magnetorquers C5 in parallel (1×4) with reaction wheels.

Although, the detumbling capabilities of the planar coils are better as compared to the torque rods but the current consumption and the corresponding power losses of planar coils are three times higher. However, it should be noted that when the reaction wheels are turned on during the detumbling phase, the magnetorquers are turned off until the point where the reaction wheels are saturated and magnetorquers are excited again to desaturate the reaction wheels. For fair comparison of the combined schemes, we left the magnetorquers in ON state after actuating reaction wheels to observe the power losses as a function of current consumptions. The plots of current drawn show that the faster is the detumbling rate the higher is the current drawn and the corresponding power consumption. It is evident that the magnetorquer rods offer better torque to power dissipation ratio as compared to the planar coils but they are bulky and occupy significant space on the spacecraft while the planar coils are embedded inside the multilayered solar panel PCB.

The preceding analyses have focused solely on single-layered planar coils. However, it is important to note that the advantage of planar coils lies in their inherent reconfigurability concept. With the capability to embed numerous coils within a multi-layered PCB, the flexibility and versatility of the design are significantly expanded, as shown in Figs. 6, 8 and 9. The power consumption of a series configuration (1×4) is lower compared to a single coil configuration, while still generating the same magnetic moment. This lower power consumption results in lower temperature gradients within the system. For instance, let's consider the case of a linearly increasing C2. In this scenario, the temperature rise of a single coil (1×1) is measured at 38.5°C, whereas the series configuration (4×1) experiences a temperature rise of 28.34°C. Thus, the series configuration demonstrates a significantly lower temperature increase. By incorporating multiple coils within the inner layers of the PCB, the number of coils can be tailored based on the specific power consumption and torque generation requirements of the mission. This flexibility allows for better adaptation to mission needs and efficient thermal management. The hybrid arrangement (2×2) of C2 generates a torque of 10.2 μNm with power dissipation of 1.26 W, showing superior performance with reasonable power consumption. It is important to note that the parallel-only arrangement (1×4) detumbles the spacecraft almost with the same speed as the combined scheme of single coil with reaction wheels, but parallel-only arrangement (1×4) generates higher temperatures embedded inside the PCB.

As the superiority of proposed coils with better detumbling rates has been established, steady state thermal analysis for single axis attitude maneuvers is required to compare the torque to power dissipation ratios with the embedded designs in available published literature. The temperatures of all the coils configuration are within thermal limits except for the parallel arrangement as tabulated in Table 1. The configuration with a parallel setup (1×4)

of the monotonously increasing C2 coil demonstrates the highest torque output of 20.4 μNm . However, this particular configuration leads to a higher temperature of 71.07°C having a power dissipation of 5.1 W. The parallel arrangement (1×4) detumbles the spacecraft within a timeframe of 50 mins. However, it is crucial to consider the potential impact of maintaining the magnetorquers' excitation for such a prolonged duration, as the resulting internal temperatures of 71.07°C could potentially impair the performance of other subsystems. Hence, the authors recommend the use of parallel arrangement (1×4) only for basic attitude maneuvers after the satellite is detumbled. Reaction wheels combined with single layered coils provide the most suitable solution for addressing high body rates in higher form factor spacecraft due to their effective power dissipation and low heat generation. To address the potential challenges posed by temperature rise, it is advisable to mitigate the issue by employing lower voltages, specifically through the implementation of a 3.3 V input range power bus design. Additionally, it should be noted that the configuration of coils (1×4 , 2×2 , 1×4) using the onboard computer can be modified as needed, allowing for flexibility and adaptability in the system. For instance, in cases where the spacecraft experiences excessively high angular rates during detumbling, a controlled approach can be employed by using the parallel configuration (1×4) initially and sequentially transitioning to the hybrid configuration (2×2) as needed. This strategy allows for better thermal management while effectively addressing the high angular rates during the detumbling process.

Analysis of Figs. 8–9 reveals crucial factors impacting the operational efficiency of coils in 3U CubeSats. Notably, the 3U CubeSat exhibits considerably faster attainment of steady state angular rates compared to the 6U CubeSat, despite an equal number of embedded layers. This unequivocally demonstrates that as the spacecraft dimensions increase in higher cube units, there is a corresponding increase in the required torque to effectively dampen the body rates. Consequently, a plausible approach to achieving faster detumbling rates involves augmenting the number of embedded magnetorquer layers within the PCB. If faster detumbling rates are of utmost importance, maximizing torque generation through an increased number of embedded layers ($n \times m$) within the PCB can significantly enhance the spacecraft's ability to reach a steady state proficiently. Regardless of the specific asymmetric magnetorquer configuration, the resulting steady state angular rate closely approximates $\vec{\omega}_{B/I} = [0, 0, 0]^T$ rad/s. The magnetorquer designs that rely on symmetrical planar windings have been found to be ineffective in detumbling CubeSats under similar initial angular velocity conditions. This signifies that conventional symmetric embedded magnetorquers cannot effectively detumble the CubeSat when it experiences high angular rates. Such embedded magnetorquers are more suitable for attitude maneuverability rather than detumbling purposes. The detumbling process of higher

Table 1

Planar magnetorquers' performance parameters for different possible arrangements compared with the commercial state-of-the-art.

Attitude control system		Dimensions (mm)	Mass (g)	Price (\$)	T (°C)	τ (μ Nm)	$\frac{\tau}{P}$
Commercial designs	Gomspace Nanopower (P110) (GomSpace, n.d.)	No space occupation	57	PCB inner layers	25.9	1.9	0.82
	Magnetorquer rod (NCTR-M012) (NCTR-M012, n.d.)	94 × 15 × 13	50	2000	25.7	60	66
	Reaction wheels (RWp015) (RWp 015., 2023)	42 × 42 × 19	130	9000	25.4	4000	5000
	MT01 compact Magnetorquer (Ecuadorian Space Agency, 2023)	50 × 50	7.5	1260	25.8	5.36	5.36
Embedded planar magnetorquer	C2 in series (4 × 1)	No space occupation	81.2	PCB inner layers	28.34	5.09	14.57
	C2 in parallel (1 × 4)	No space occupation	81.2	PCB inner layers	71.06	20.40	3.32
	C2 in hybrid (2 × 2)	No space occupation	81.2	PCB inner layers	38.51	10.2	8.51
	C5 in series (4 × 1)	No space occupation	81.2	PCB inner layers	27.42	4.05	21.21
	C5 in parallel (1 × 4)	No space occupation	81.2	PCB inner layers	57.51	16.22	4.81
	C5 in hybrid (2 × 2)	No space occupation	81.2	PCB inner layers	34.32	8.01	10.12

form factor 6U CubeSats poses challenges for single coil asymmetric magnetorquer topologies, as evidenced in Fig. 7. This demonstrates the obvious requirement for multilayered configurations ($n \times m$) as the spacecraft's dimensions and inertia increase. It is crucial to note that in the event of a worst-case scenario where the reaction wheels fail, the reconfigurable arrangements of an asymmetric layout specific to the application can be utilized effectively to detumble the spacecraft. Malfunctions in reaction wheels commonly occur in small satellites and often result in failures of the attitude control subsystem. Faults in reaction wheels include bearing failures, imbalance, mechanical wear, and motor faults. These issues can lead to reduced performance, vibrations, decreased efficiency, and potential system failures. It is crucial to note that in the event of a worst-case scenario where the reaction wheels fail, the reconfigurable arrangements ($n \times m$) of an asymmetric layout specific to the application can be utilized effectively to detumble the spacecraft.

The proposed embedded planar magnetorquer configurations demonstrate a direct relationship between torque and detumbling rate, where higher torque values correspond to faster detumbling rates. The C2 in parallel (1 × 4) configuration exhibits the highest torque value of 20.40 μ Nm, resulting in the fastest detumbling rate, but with a relatively low torque-to-power-dissipation ratio of 3.32 and a high temperature of 71.06 °C. The C2 in hybrid (2 × 2) and C5 in parallel (1 × 4) configurations follow, with torque values of 10.2 μ Nm and 16.22 μ Nm, respectively, and torque-to-power-dissipation ratios of 8.51 and 4.81, respectively, and temperatures of 38.51 °C and 57.51 °C, respectively. The C2 in series (4 × 1) configuration shows a relatively lower torque value, leading to a slower detumbling rate, but with a high torque-to-power-dissipation ratio of 14.57 and a low temperature of 28.34 °C. The C5 in series (4 × 1) configuration has the lowest torque value of 4.05 μ Nm, resulting in the slowest detumbling rate, but with an impressive torque-to-power-dissipation ratio of 21.21 and a low temperature of

27.42 °C, indicating high efficiency and reduced heat generation. Compared to commercial attitude control systems, embedded planar magnetorquers offer several advantages. They have a compact design, low mass, and are affordable. For example, the Gomspace Nanopower (P110) has a compact design and low mass (57 g), but its torque output (1.9 μ Nm) and torque-to-power ratio (0.82) are relatively low. The magnetorquer rod (NCTR-M012) provides moderate torque output (60 μ Nm) and torque-to-power ratio (66) at a higher price point (\$2000). The MT01 compact magnetorquer offers compact dimension with a mass of 7.5 g, a torque output of 5.6 μ Nm, a torque-to-power ratio of 5.36 at a price of \$1260. In contrast, Reaction wheels (RWp015) offer substantial torque output (4000 μ Nm) and torque-to-power ratio (5000) at a higher price (\$9000).

5. Conclusion

The successful detumbling of high form factor satellites at high angular velocities using off-the-shelf reaction wheels and copper layers within the PCB is undeniably a remarkable achievement. This disruptive approach signifies significant progress towards achieving efficient and effective detumbling of CubeSats while simultaneously preserving valuable spacecraft space. By integrating planar coils with diverse geometries alongside reaction wheels, this combined scheme presents promising solutions for attaining precise control of CubeSat attitudes in space missions. Furthermore, the reconfigurable design introduced in this study provides a distinct advantage by offering the freedom to generate any desired magnetic moment while effectively managing power dissipation and heat generation within the PCB. This inherent flexibility allows for precise customization of the magnetic properties to align with the specific requirements of the mission or application. Moreover, the reconfigurable nature of the design ensures adaptability and scalability, paving the way for potential future improvements in reusability and enhancements in overall

performance. The detumbling rates achieved by the proposed design were compared with both commercial designs and relevant literature, exhibiting a superior performance. Future research could investigate the benefits of combining embedded asymmetric magnetorquers with miniaturized propulsion thrusters for attitude control, enabling CubeSats to perform complex maneuvers and missions.

Conflicts of Interest

The authors declare no conflict of interest.

CRedit authorship contribution statement

Shoaib Ahmed Khan: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zou Tao:** Writing – review & editing, Validation, Supervision, Investigation. **Shah Fahad:** Writing – review & editing, Validation, Resources. **Muhammad Salman:** Writing – review & editing, Software. **Mustafa Tahir:** Visualization, Validation, Software, Methodology. **Anwar Ali:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization.

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